Benjamin Vaccaro

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Embedded systems, RF, and software

Education

Millis High School, Millis, MA — Expected June 2027

National Honor Society · AP Scholar with Honor · Stanley Z. Koplik Certificate of Mastery

Coursework: AP Physics C: Mechanics and E&M (VHS Learning), AP Calculus AB, Engineer Your World Honors (dual enrollment), AP Statistics (VHS Learning), AP Computer Science Principles

Experience

MIT Lincoln Laboratory, Beaver Works Summer Institute: UAS Synthetic Aperture Radar — Jul – Aug 2026

Engineering Award recipient · selected as 1 of 21 from 400+ applicants

- Built the web interface and helped integrate the data pipeline for a five-person team's drone-mounted radar imaging system, delivered for a live final demonstration.
- Improved the efficiency of the team's back projection imaging algorithm; worked across firmware, signal processing, and flight operations on timestamp alignment, per-pulse positioning, and payload integration.

Synthica, Software Engineering and DevOps Intern — May – Aug 2026

- Built infrastructure and automation for research publication workflows, owning pipeline components end to end.

Town of Millis, Digital Operations Volunteer — 2025 – Present

Run online donation systems for the Council on Aging and food pantry; maintain municipal and historic-trail websites.

Projects

- **intraweb** (Rust): Zero-config local network in one static binary, no internet or server. Nodes find each other over mDNS and UDP broadcast; signed mail, end-to-end-encrypted channels (Ed25519 / X25519), resumable file sharing, and ESP32 firmware that hosts the network without a router. 257 tests.
- **stdnt.xyz** (live): Searchable directory of 650+ free resources for students, including scholarships, STEM programs, competitions, and developer tools, with deadline calendar feeds and an on-device eligibility quiz. Built with All The Same Organization; open source with community submissions.
- **Backoffice** (Go, SQLite): Inventory system for a whole workshop in one binary: parts with photos and drawers, KiCad BOM import, wiring diagrams, and G-code sent to Klipper or OctoPrint. Runs on a Raspberry Pi.
- **LoRa Mesh Network:** Neighborhood messaging network on custom PCBs I designed and assembled, with multi-hop routing so nodes relay traffic without a central server or cell tower.
- **MBTA Tracking Wearable:** 1st place overall, MIT Blueprint. Wrist-worn transit tracker built in under 24 hours from the firmware up: embedded C, live API polling over WiFi, OLED display.

Research

AlphaFold2 Accuracy on Viral vs. Cellular Proteins — Mentor: Dr. Blair Paul, Marine Biological Laboratory

Compared 233 solved structures with TM-align: AlphaFold2 is significantly less accurate on viral proteins (median TM-score 0.927 vs. 0.970, $p = 1.3\times10^{-8}$), independent of intrinsic disorder. An ESMFold control showed multiple sequence alignments partly rescue viral predictions. Paper in preparation.

Leadership and Service

- **U.S. Naval Sea Cadet Corps:** Command Leading Petty Officer, the senior cadet billet in the Massachusetts Bay Division. Write the Plan of the Day and run field training; unit earned a perfect score on its 2025 national inspection.
- **Hack Club:** Boston Ambassador. Organized 2 hackathons and 3 workshops for high schoolers; co-organizing a fall 2026 hackathon with Artisan's Asylum.
- **SeaPerch Underwater Robotics:** Team Captain. Took the team from unranked to Top 50 nationally; teach soldering and circuit design to younger cadets.
- **Boy Scouts of America, Troop 15:** Eagle Scout candidate. Led historic trail signage at 10 sites across Millis. Senior Patrol Leader of National Youth Leadership Training (16 staff, 60 participants).

Skills

Programming: Python, C/C++, Rust, Go, JavaScript (Node.js), SQL, Bash
Hardware: PCB design (KiCad), firmware for ESP32 / ARM / Arduino, RF and antenna design, soldering, Fusion 360
Systems: Linux / NixOS, Git, REST APIs, SQLite, Docker, self-hosted services, NumPy, SciPy, Pandas
Languages: English and Spanish (bilingual)

Honors

SAR Leadership Award · International Research Olympiad Semifinalist · Computer Science Student of the Year